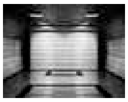

Driverless garage
Daimler and Bosch have created a driverless parking garage for the Mercedes-Benz Museum in Stuttgart. **http://dgit.in/nodrivergarage**

Roomba home data
Roomba could be selling data collected by its devices to home-automation companies. **http://dgit.in/iroombaspy**

home-delivered pizza faster than light. It's really a misnomer to even use the word "teleportation", but hey, physicists need to feel cool too. The only thing being teleported is information, and the hope is that (if it works) it *might* revolutionise the concept of the internet, communication and computation in general. How? Let's find out...

THE HOW

Quantum physics is not intuitive to us humans because it works at scales pretty much unimaginable to us. It turns out that the quantum scale (about a quadrillionth of a metre, or 15 zeros in the denominator) deals with strange, abstract properties like 'spin' for electrons, and 'polarisation' for photons of light. While they're abstract to us in the everyday world, these properties can be, and have been, measured, multiple times, in hundreds of different experiments. This stuff isn't made up.

Spin and polarisation are interesting to physicists for a whole array of reasons that it would take a few years to cover. What's key here is that an 'up' or 'down' spin, or a 'vertical' or 'horizontal' polarisation – which are measurable states – are kind of analogous to a 0 or 1, a bit. These cool, quantum bits are also called qubits, and can be both 1 and 0 at the same time!

Here's the final piece in the puzzle – the behaviour of these quantum bits (just electrons or photons) is extremely well-understood mathematically. They follow the well-defined rules of quantum mechanics to a T. And quantum mechanics uses all this math to explain behaviours like 'entanglement' (where the states of multiple qubits are fundamentally related to each other in a mathematically inseparable way), 'interference' (where a qubit can cross its own trajectory and 'interfere' with its own path) and a host of other jargon.

Don't forget, all this stuff is verified with experiments involving crystals and mirrors. These supposedly simple apparatuses can actually manipulate quantum states in a variety of ways, even if they don't make intuitive sense. Quantum physicists have learnt to turn off their intuition and follow the evidence instead.

Well, what good is all this math and physics? In 1993, Charles Bennett proposed that entanglement could be used to 'teleport' the quantum state of a qubit to another entangled qubit across a large distance without moving either particle. More than 20 years on, this has actually been done. Multiple times. In fact, most recently, a Chinese satellite was successfully communicated an entangled state from a particle on the ground, 1,400 kilometres away. This is more than eight times the previous record. This is mind-blowing, a proof of concept of an idea thought pretty much impossible such a short time ago.

However, another reality check is in order: over 32 nights of experimentation, only about 900 pairs of qubits showed verifiable teleportation. This is because working with light is tricky, and there are just too many variables. From the slightest change in atmospheric density to the smallest vibration, pretty much anything can disturb the experiment – they are, after all, dealing with reading spins on individual electrons!

So why is quantum teleportation even considered useful? Let's break it down:

SPEED

Without quantum physics, the fastest we can beam information from one place to another is at the speed of light. On the ground, most information is transmitted by cables at about two thirds of that speed. The best fibre optic cables operate at 99% of the speed of light, and cost a bomb. Quantum teleportation of information would be able to bypass all such restrictions. and information is transferred instantly, and thus the speed is, well, infinite!

When we say big brains, we're not kidding around... Pictured: Einstein, the Curies, Schrodinger, Heisenberg, and just a few other Nobel Prize winners